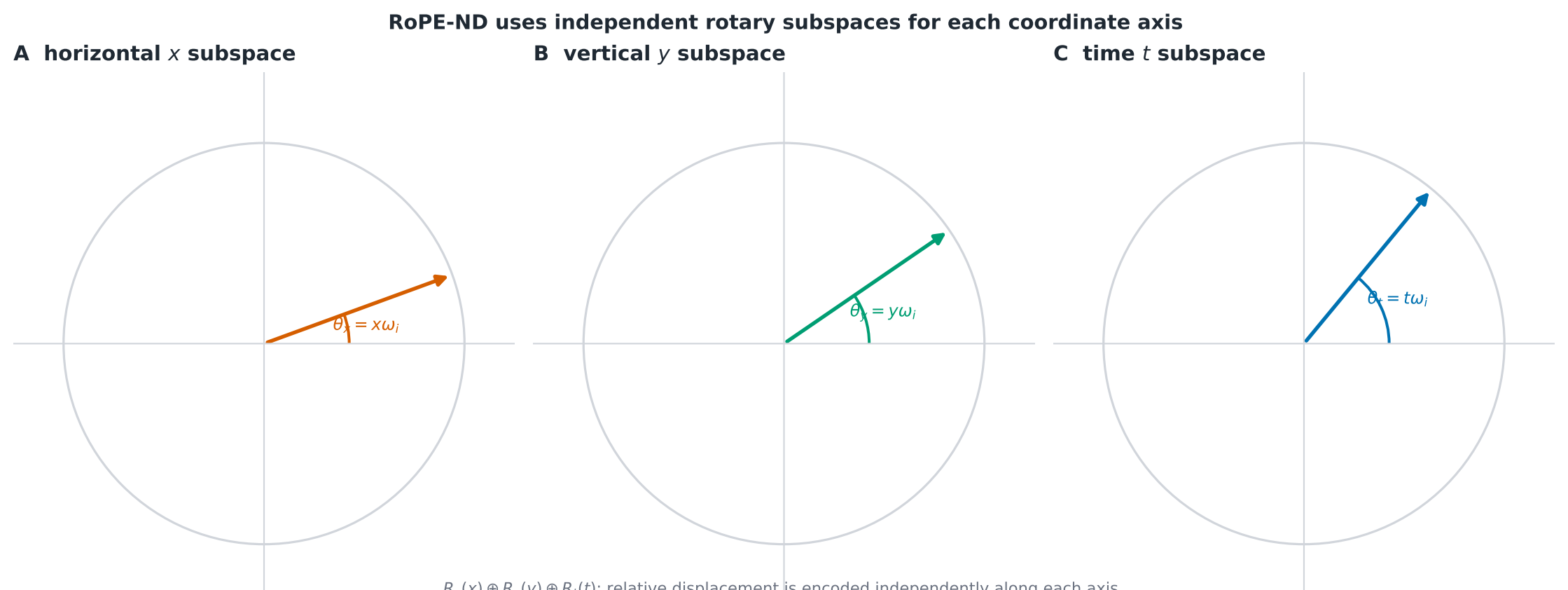
RoPE-ND uses independent rotary subspaces for each coordinate axis
A horizontal x subspace
$\theta_x = x\omega_i$
B vertical y subspace
$\theta_y = y\omega_i$
C time t subspace
$\theta_t = t\omega_i$
$R_x(x) \oplus R_y(y) \oplus R_t(t)$: relative displacement is encoded independently along each axis.